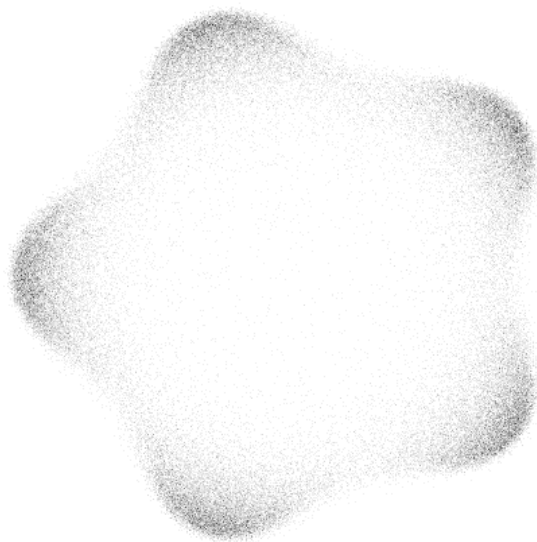


ASYMPTOTICS OF PARTITION FUNCTIONS IN RANDOM MATRIX THEORY

Annual Scientific Report for the Masters project, fomented by Fundação de
Amparo à Pesquisa do Estado de São Paulo.

Project FAPESP #2024/06997-1

Supervisor: Guilherme L. F. Silva



São Carlos, March 9, 2026

Project's General Information

- Project Title:

Asymptotics of partition functions in random matrix theory

- Supervising Researcher:

Guilherme L. F. Silva

- Institution:

**Instituto de Ciências Matemáticas e de Computação (ICMC) da
Universidade de São Paulo**

- Research team:

Guilherme L. F. Silva

João Victor Alcantara Pimenta

- Research project number:

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- Project effect dates:

20/04/2025 a 20/03/2026

- Dates covered by this report:

01/08/2024 a 31/07/2026

Resumo

The theory of Random Matrices finds its way in a diverse range of areas of Mathematics and Physics, and in particular to statistical mechanics. In the study of the spectra of eigenvalues of some random matrices ensembles, a physical analogy regarding thermodynamical systems becomes evident and the use of physical motivations in its study is of great importance. An example of such motivation is the study of the partition function, which holds in thermodynamics a great deal of information on the system at hand. Recent developments have been intensely unraveling new critical phenomena in random matrix theory, through a deep understanding of asymptotic expansions of the associated partition functions in the thermodynamic limit. The main goal of this work is to understand these developments in the context of normal random matrices, their relevance and implications in the theory of random matrices and beyond.

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1 Period Achievements

1.1 DEVELOPED ACTIVITIES

The following list summarizes the activities for the months covered by the report.

1. **Completion of the Courses Necessary Credits:** It was completed - in the second semester of 2025 - the necessary courses to fulfill the credits requirements for the Masters course in the Institution;
2. **Publication of papers:** in the time covered by this report two papers were published where the process beneficiary is a co-author. “A Central Limit Theorem for intransitive dice” published in *ALEA, Lat. Am. J. Probab. Math. Stat. (v22-46)*, 2025, with DOI 10.30757. “Simulating Simple Random Walks With a Deck of Cards” published in *Brazilian Journal of Probability and Statistics (25-BJPS635)*, 2025, with DOI 10.1214.
3. **Participation in Events:** Many events have been attended in the period covered by this report. It was presented a poster (B) at CLAPEM - Latin American Congress of Probability and Mathematical Statistics held in Montevideo - Uruguay in March, 2026. Money from the research fund was used to take part in this event. A poster (A) was also presented at the “XV Workshop de Teses e Dissertações em Matemática” held in São Carlos - Brazil in October, 2025. He took part in the “Fourier Analysis & Beyond I” held in IMPA, Rio de Janeiro from June 30th to July 04th, 2025. He also took part in the “35^o Colóquio Brasileiro de Matemática” held IMPA, Rio de Janeiro from July 27th to August 1st, 2025. The student also took part in an extension project called SigmaCamp <https://sigmacamp.org/> - and international advanced scientific camp for young students. He took part in the ETAPA SigmaCAMP and the SigmaFestival where he taught classes on mathematical and physical subjects besides doing administrative tasks along a team of international researchers from all sciences and institutions.
4. **Continuous study of theory and meaningful literature:** As the natural way to research, we gave continuation to the reading material in the theory and started focusing - in the second semester of 2025 - in the main research article [1] we will work on the remaining months;
5. **Masters Thesis:** It was begun the process of organizing and writing what will become the final product of the masters project, the thesis.

1.2 SCHEDULE

This M.D. research proposal started in August 2024 and is running for the usual time for the Masters Program, that is, 24 months. The project was originally divided for execution in the following schedule.

Steps	Quarters							
	1	2	3	4	5	6	7	8
1. Master Courses	x	x	x	x	x	x		
2. Concepts			x	x	x	x		
3. Main Article				x	x	x	x	
4. Exploration					x	x	x	
5. Masther Thesis					x	x	x	x

Table 1.1: Activities originally planned. Red color indicates the focus of the quarter that may be divided between activities. Green indicates the planning was accomplished.

Naturally then, until the end of the project the student should be finalizing the study of the main article and writing the final thesis as he explores the possibilities of the research.

1.3 MASTERS COURSES

During the Masters the student took part in the following courses.

Course	Code	Concept	Semester
Variedades Diferenciáveis	SMA5781-21/3	A	2 - 2024
Equações Diferenciais Ordinárias	SMA5802-17/2	B	2 - 2024
Algebra	SMA5738-17/4	A	1 - 2025
Medida e Integração	SMA5801-13/4	C	1 - 2025
Introdução a Mecânica Celeste	SMA5998-1/1	A	2 - 2025
Preparação Pedagógica	SMA5839-19/8	A	2 - 2025
Treinamento Profissional e Acadêmico	SMA5976-2/1	A	2 - 2025

1.4 RESEARCH OVERVIEW

Random Matrices can be thought of in two ways: As a measurable function from a probability space into a set of matrices or, more concretely, as a matrix whose entries are random variables with some joint distribution. In some cases there is an explicit density formula for the eigenvalues of such a random matrix. For some Random Matrices, with the help of some mathematical tools such as orthogonal polynomials one can express the equilibrium measures for the eigenvalues asymptotically. There are many instances of “models” of random matrices, each one with its symmetries and own probability density. Those are what we call *ensembles*.

Borrowing some concepts from physics, an ensemble can be thought as a virtual collection of all possible states that a system can be in, with more probable states appearing more often in the collection. There is an analogy to be made with ensembles of thermodynamic: When considering sets of matrices with equal probability - in some sense to be more clear soon - we get the Gibbs weight for the probability distribution as we would in the physical *Canonical Ensemble*, thought of as a system of particles with constant temperature. More concretely, in the context of Random Matrix Theory, we have the definition.

Definição 1.4.1 *A random matrix ensemble $\mathcal{E} = (\mathcal{S}, \mathcal{P})$ is a set of matrices $\mathcal{S}$ of fixed size equipped with a probability density function (p.d.f.) $\mathcal{P} : \mathcal{S} \rightarrow [0, \infty)$. If we say that a matrix $\hat{X}$ is drawn from the random ensemble $\mathcal{E}$, we are saying that it is a random variable assuming values in $\mathcal{S}$ with p.d.f. $\mathcal{P}$.*

In the research proposed here, we will focus on the study of *normal random matrices*, which consists of the space of normal matrices, equipped with a proper probability distribution. In the beginning of its book, Feynman states [2] that the key principle of the statistical mechanics is that the probability that a system attains a state with energy E , in equilibrium, is given by a function $\frac{1}{Z} e^{-\frac{E}{kT}}$ where Z is the partition function. In a more general sense there is a relation between the partition function and the free energy of the system, that, in itself, relates to the entropy. Knowing well the partition function is a way to describe with great detail the macro-properties of such a system.

There is a great effort in the community of random matrix theory to study expansions of the partition function of various classes of Random Matrix. Recently, some advance has been made in the works of Sug-Soo Byun et al. [1]. The work they done was on Random Normal Matrix and Planar Ensembles (with Determinantal and Pfaffian structures, respectively), which present partition functions

$$Z_N^{(\beta)} := \int_{\mathbb{C}^N} \prod_{j>k=1}^N |z_j - z_k|^\beta \prod_{j=1}^N e^{-\frac{\beta N}{2} Q(z_j)} dA(z_j) \quad (1.4.1)$$

where N is the size of the matrix, β is relative to the generic entries dimension and $dA(z) := d^2z/\pi$ is the area measure. Here, $Q : \mathbb{C} \rightarrow \mathbb{R}$ is called the confining potential and satisfies some suitable conditions. In their work, using concepts of orthogonal polynomials and integral asymptotic by Laplace method, they derived large- N asymptotic expansions up to the $O(1)$ -terms for $Z_{N,\beta}$ when V is a radially symmetric potential, in the form

$$Z_{N,\beta=2} = -I^V N^2 + \frac{1}{2} N \log N + E^V N + G \log N + F^V + o(1), \quad N \rightarrow \infty.$$

In such expansion, for a large class of potentials V the first term I^Q was known for a long time, being given by the energy of the associated equilibrium measure. The coefficient $1/2$ of the order $N \log N$, as well as the entropy term E^V , were known from recent work of Leblé and Serfaty [3], also for rather large classes of V . The remaining terms in this expansion are new, and they are computed exploring the radially symmetry imposed on

the potential. Strikingly, the term G appears to be universal in V , depending solely on the connectivity of the limiting spectrum of eigenvalues. More precisely, they noticed that G depends on whether the limiting spectrum is an annulus or a disc which, surprisingly, seems to not have been observed in the vast previous literature on the matter.

As was said, the study of the partition function is a major way to describe thermodynamic systems. With these new developments it is expected that many systems could be better understood and described by the asymptotic given. In that way this poses a great opportunity for development in the field. More than that, by the great connection that the field of random matrices holds with many other fields is possible that by the study of this work some suggestions on the behavior of many other related problems become plausible. Especially, we will be interested in external field model.

2 Scientific Events

Many events have been attended in the period covered by this report. It was presented a poster (B) at **CLAPEM** - Latin American Congress of Probability and Mathematical Statistics held in Montevideo - Uruguay in March, 2026. Money from the research found was used to take part in this event. A poster (A) was also presented at the “XV Workshop de Teses e Dissertações em Matemática” (**WPGMAT**) held in São Carlos - Brazil in October, 2025. He took part in the “Fourier Analysis & Beyond I” (**FAB**) held in IMPA, Rio de Janeiro from June 30th to July 04th, 2025. He also took part in the “35^o Colóquio Brasileiro de Matemática” (**CBM**) held IMPA, Rio de Janeiro from July 27th to August 1st, 2025. The student also took part in an extension project called SigmaCamp <https://sigmacamp.org/> - and international advanced scientific camp for young students. He took part in the ETAPA SigmaCAMP and the SigmaFestival where he taught classes on mathematical and physical subjects besides doing administrative tasks along a team of international researchers from all sciences and institutions.

Event	Institution	Date	Modality	Presentation	Research founds
FAB	IMPA	30/06-04/07/2025	Presential	Participant	No
CMB	IMPA	27/07-01/08/25	Presential	Participant	No
WPGMAT	ICMC-USP	01/08/2025	Presential	Poster - Oral	No
CLAPEM	SLAPEM	01-06/03/23	Presential	Poster - Oral	Yes

3 Other prepared or submitted works

During the period covered by this report two papers were published. The first one was “A Central Limit Theorem for intransitive dice” (C), published in ALEA, Lat. Am. J. Probab. Math. Stat, 2025, with DOI 10.30757. The second was “Simulating Simple Random Walks With a Deck of Cards” (D), published in Brazilian Journal of Probability and Statistics, 2025, with DOI 10.1214. Both are presented in the Appendix.

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A Poster Workshop

Poster presented in the “XV Workshop de Teses e Dissertações em Matemática” in ICMC. São Carlos, Brazil. 2025.

Asymptotics of Partition Functions on Coulomb Gases

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XV Workshop de Teses e Dissertações em Matemática

The theory of random matrices finds its way in a diverse range of areas of Mathematics and Physics, and in particular in statistical mechanics. In the study of the spectra of eigenvalues of some random matrix ensembles, a physical analogy regarding thermodynamical systems becomes evident and the use of physical motivations in its study is of great importance. To better understand these systems, we study a known concept for physicists - the so-called partition functions for Coulomb Gases. These objects are of fundamental importance in the pursuit of understanding models of random matrices - thought of as particle systems composed of eigenvalues. In particular, we are concerned about random matrices of large size. Our goal is then to have a good understanding of the asymptotic of the partition function for the given large system. Recently, asymptotics for some classes of potentials for two dimensional Coulomb Gases have been calculated using a novel method based on asymptotics of orthogonal polynomials, a much more straightforward approach that enables us to explore the partition functions for other systems of interest. In light of this, we will properly understand this result and look for possible models where we can reapply the technique for new results. We start our study with an important class of examples which have its asymptotics expressed in terms of Hermite functions. To study them, we use their integral representation, going through a better understanding of classical methods for asymptotics of integrals, such as the Watson Lemma, Laplace Method and Steepest Descent, to name a few. Besides characterizing these asymptotics, we understand its place in the theory of random matrix and search for new results the technique could provide.

Density Function of Matrix Models

Invariant ensembles of random matrices, that we are interested in, are the ones in which the probability distribution is expressed as

$$p(\vec{M})d\vec{M} = \frac{1}{Z_N} e^{-\text{Tr}(V(\vec{M}))} |\Delta(\Lambda)|^2 d\Lambda dU,$$

for any polynomial potential $V: \mathbb{R} \rightarrow \mathbb{R}$ such that $V(x) \rightarrow \infty$ sufficiently rapidly as $|x| \rightarrow \infty$. Here, we also considered Weyl's formula to express $d\vec{M} = |\Delta(\Lambda)|^2 d\Lambda dU$. We note that $|\Delta(\Lambda)|$ is the jacobian expressed as a Vandermonde determinant, $d\Lambda$ is the Lebesgue measure on $\mathbb{R}^N$, dU is the Haar measure on the set of invertible matrices and β is the inverse temperature, which we take to be 2 for the rest of this work. As this ensemble is invariant, we can express the jpdf of the random matrix as a function on the eigenvalues

$$p_{\text{ord}}(\lambda_1, \lambda_2, \dots, \lambda_N) = \frac{1}{Z_{N,\beta}^{(\text{ord})}} \prod_{i=1}^N e^{-V(\lambda_i)} \prod_{i < j} |\lambda_i - \lambda_j|^2. \quad (1)$$

This has an interesting interpretation as a Coulomb gas. The measure models an interacting gas of indistinguishable particles under the influence of an external potential at positions $\lambda_1, \lambda_2, \dots, \lambda_N \in \mathbb{C}$ in an external ambient of two dimensions.

The Kernel Representation

There is a further manipulation we could do on the expression given by 1. Namely, for $\beta = 2$

$$\begin{aligned} p_{\text{ord}}(\lambda_1, \lambda_2, \dots, \lambda_N) &= \frac{1}{Z_{N,\beta}^{(\text{ord})}} \left[\det(m_i^{j-1})_{i,j=1,\dots,N}^2 \right] \prod_{i=1}^N e^{V(\lambda_i)} \\ &= \frac{1}{Z_{N,\beta}^{(\text{ord})}} \left[\det(p_{j-1}(\lambda_i))_{i,j=1,\dots,N}^2 \right] \prod_{i=1}^N e^{V(\lambda_i)} \\ &= \frac{1}{Z_{N,\beta}^{(\text{ord})}} \det(K_N(\lambda_i, \lambda_j)) \prod_{i=1}^N e^{V(\lambda_i)}, \end{aligned}$$

where $K_N(x, y) = \sum_{j=0}^{N-1} p_j(x)p_j(y)$. If we choose p_j to be the Hermite polynomial, by algebraic properties

$$\begin{aligned} \dot{K}_N(x, y) &= e^{-\frac{V(x)}{2}} e^{-\frac{V(y)}{2}} K_N(x, y) \\ &= e^{-\frac{V(x)+V(y)}{2}} \frac{H_N(x)H_{N-1}(y) - H_{N-1}(x)H_N(y)}{x-y}. \end{aligned} \quad (2)$$

This motivate the study of asymptotics of orthogonal polynomials, especially the Hermite polynomial.

Asymptotics Methods

We now focus on Hermite polynomials. It admits the integral representation

$$H_n(x) = \frac{n!}{2\pi i} \int_{\Gamma} z^{-n-1} e^{2xz-z^2} dz =: c_{\lambda} \int_{\Gamma} g(s) e^{-\lambda f(s)} ds = \phi(\lambda) \quad (3)$$

where Γ is a circle around the origin for the Hermite polynomial.

Theorem 1 (Steepest Descent) Take γ a smooth contour. Assume that the exponent $f = u + iv$ is an analytic function on a neighborhood of Γ that satisfies $v(z) \equiv v_0, z \in \gamma$. In addition, suppose that u has a unique minimum at an interior point of Γ , say on $z_0 \in \gamma$, with $f''(z_0) \neq 0$, and also that g is analytic on a neighborhood of γ and satisfies $\int_{\gamma} |g(z)||dz| < \infty$. Then ϕ has a leading asymptotic given by

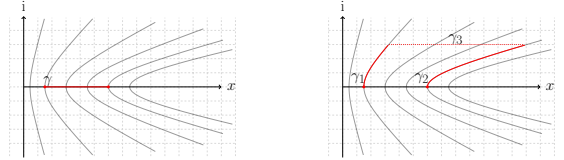
$$\phi(\lambda) = e^{-\lambda f(z_0)} \left(\frac{\sqrt{2\pi} g(z_0) e^{i\theta(z_0)}}{\lambda^{1/2} \sqrt{|f''(z_0)|}} + \mathcal{O}(\lambda^{-3/2}) \right), \quad \lambda \rightarrow +\infty, \quad (4)$$

where $\theta(z_0)$ is the angle of the oriented tangent of γ at z_0 .

The assumption of analytic functions, allows the deformation of the contour. To choose an appropriate contour, note that the gradient for the imaginary and real part are orthogonal at any point, i.e.

$$\langle \nabla u(z), \nabla v(z) \rangle = u_x(z)v_x(z) + u_y(z)v_y(z) = u_x(z)v_x(z) - v_x(z)u_x(z) = 0.$$

Taking the critical points of f we may find a path passing through where the imaginary part is constant, this will follow the *Steepest Descent* path of the real part.



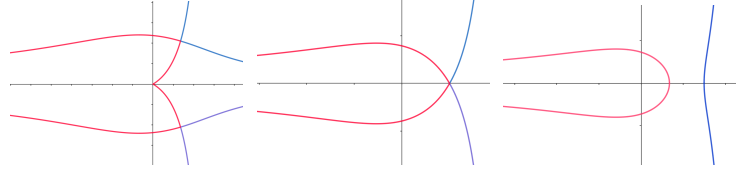
The problem reduces to the evaluation of real integrals. We note that, in the limit, the integrand accumulates around the critical points. Theorem 1 gives an estimative for the integral, given the critical point.

Hermite Polynomials

To use the result on the Hermite polynomials, as in 3, we first consider the scaling $x \mapsto \sqrt{n}\alpha$ and $z \mapsto \sqrt{n}\beta$

$$H_n(\sqrt{n}\alpha) = C_n^0 \int_{\Gamma} e^{-n(\phi_n(\beta))} \beta^{-1} d\beta, \quad H_{n-1}(\sqrt{n}\alpha) = C_n^1 \int_{\Gamma} e^{-n(\phi_n(\beta))} d\beta,$$

where $\phi_n(\beta) = -2\alpha\beta + \beta^2 + \log(\beta)$. We determine the critical points $\beta_c^{\pm} = \frac{\alpha \pm \sqrt{\alpha^2 - 2}}{2}$ as solutions to $\phi'_n(\beta) = 0$. Because of the critical point behaviour, we divide the problem into three cases: $0 < \alpha < \sqrt{2}$, $\alpha = \sqrt{2}$ and $\alpha > \sqrt{2}$ and solve for $\text{Im}(\phi_n(z) - \phi_n(\beta_c^{\pm})) = 0$. The solution to this equation are the paths shown below



We now apply Laplace Method for each deformation on the determined paths of *Steepest Descent*, determining the asymptotic related to each case. We show results for H_n but the same is done for H_{n-1} .

• For $0 < \alpha < \sqrt{2}$: Γ deformed to $\Gamma_d^+ \cup \Gamma_d^- \cup \Gamma_x$, $\alpha = \sqrt{2} \cos(\gamma)$ and $\beta_c^{\pm} = \sqrt{2}/2 e^{\pm i\gamma}$

$$H_n(\sqrt{2n} \cos(\gamma)) \approx \frac{n! n^{-\frac{1}{2}(n+1)}}{2 \sin(\gamma) \sqrt{\pi}} e^{-n \left(-\cos^2(\gamma) - \frac{1}{2} + \ln \frac{\sqrt{2}}{2} \right)} \cos(n\theta\gamma - \theta(\beta_c^{\pm}))$$

• For $\alpha > \sqrt{2}$: Γ deformed to Γ_{β^-} , $\alpha = \sqrt{2} \cosh(\gamma)$ and $\beta_c^{\pm} = \sqrt{2}/2 e^{\pm \gamma}$

$$H_n(\sqrt{2n} \cosh(\gamma)) \approx \frac{n! n^{-\frac{1}{2}(n+1)} e^{\gamma}}{\sqrt{2\pi} |1 - e^{-2\gamma}|} e^{-n \left(-2 \cosh(\gamma) e^{-\gamma} + \frac{1}{2} e^{-2\gamma} + \gamma + \ln \left(\frac{\sqrt{2}}{2} \right) \right)}.$$

Conclusion

Given the well-established importance of random matrices, we explore its properties. Using techniques of deformation of path of integration on complex integrals and estimatives of real integrals with critical points under some limit, we determine the asymptotics of Hermite Polynomials H_n . With this result, we can also determine asymptotics on the so-called kernel K_n . This object has great importance as it shows up in the expressions of the equilibrium measures for models of random matrices. Therefore, we can study properties of such models related to these measures. This is the next step in the project: asymptotic analysis of partition functions on Coulomb gases, an analogue of models of random matrix, for settings of interest.

Acknowledgements

We would like to thank the São Paulo Research Foundation (FAPESP) for Financial Support under Process Number 2024/06997-1. We also thank the Institute of Mathematical and Computer Sciences (ICMC) for the Institutional Support under the PPG-MAT masters program.

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B Poster CLAPEM

Research Funds were used to take part in this event, because of that I declare that: This work was presented as a poster in the XVII CLAPEM - the Latin American Congress of Probability and Mathematical Statistics which has occurred on March 2-6, 2026 in Montevideo, Uruguay.

Contextualization

Random Matrices can be thought of in two ways: As a measurable function from a probability space into a set of matrices or, more concretely, as a matrix whose entries are random variables with some joint distribution. A set $\mathcal{S}$ of such matrices with a given arbitrary p.d.f. $\mathcal{P}$ is called an ensemble. While studying eigenvectors may be of interest in some cases, eigenvalue statistics have been considerably more explored for these matrices. Hence, in addition to the p.d.f.s, we are also interested in the eigenvalue j.p.d.f. $p(\lambda_1, \lambda_2, \dots, \lambda_n)$, and the subsequent univariate eigenvalue density $p(\lambda_1; n)$.

Here, *invariant matrix ensembles* refers to ensembles which distribution can be expressed as a function of its eigenvalues. They are invariant by some appropriate group action. We are mainly interested in the subset called *Orthogonal Polynomial Ensembles* when $\beta = 2$ - that is, sets of matrices with j.p.d.f.

$$p(\lambda_1, \lambda_2, \dots, \lambda_n; \beta) = \frac{1}{Z_{n,\beta}^{(w)}} \prod_{i=1}^n w(\lambda_i) |\Delta_n(\lambda)|^\beta$$

for some continuous, non-negative, real-valued weight function $w(\lambda)$ and

$$\Delta_n(\lambda) := \det [\lambda_i^{j-1}]_{i,j=1}^n = \prod_{1 \leq i < j \leq n} (\lambda_i - \lambda_j)$$

being the Vandermonde determinant. Such ensembles have a natural interpretation as a *Coulomb Gas* - an interacting gas of indistinguishable particles under an external potential.



The weight corresponds to an external potential and the determinant to the appropriate coulombian interaction kernel. One can actually show that for these ensembles $p(\lambda; \beta)$ is given by the formula

$$p(\lambda; 2) = K_n(\lambda, \lambda) = \sqrt{w(x)w(y)} \sum_{k=0}^{N-1} \frac{1}{n_k} q_k^{(2)}(x) q_k^{(2)}(y),$$

where $\{q_j^{(\beta)}\}_{j=0}^\infty$ is a family of monic polynomials orthogonal in respect to $w(x)$, such that each $q_j^{(\beta)}(\lambda)$ is of degree j . Then, there exists finite positive constants, called norming constants, $\{n_j\}_{j=0}^\infty$ s.t.

$$\langle q_j, q_k \rangle^{(2)} = n_j \chi_{j=k}.$$

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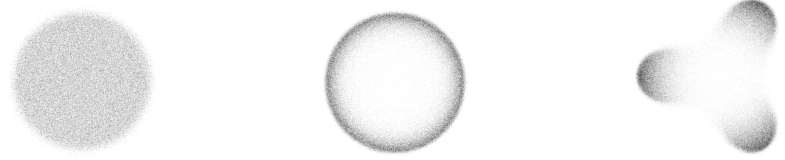
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Models of Interest

We will focus on a model related to Coulomb Gases in the complex plane, one could write explicitly its partition function, that is, the normalization for the distribution

$$Z_{N,\beta}^Q := \int_{\mathbb{C}^N} \prod_{j>k=1}^N |z_j - z_k|^\beta \prod_{j=1}^N e^{-\frac{\beta N}{2} Q(z_j)} dA(z_j).$$

where $dA(z) := d^2z/\pi$ is the area measure. This is the expression, if $\beta = 2$, for the eigenvalue distribution of the so-called **random normal matrices**. For $Q(z) = |z|^2$, for example, this would be the **complex Ginibre ensemble**, the simplest of the *Gaussian Ensembles*, arguably the most notorious class of ensembles.



We consider some potential radially symmetric $Q(z) : \mathbb{C} \rightarrow \mathbb{R}$ which satisfies some nice growth condition with $q(|z|) = Q(z)$. In a recent work [1] large- N asymptotic expansions were derived up to the $O(1)$ -terms for $\log(Z_{N,\beta}^Q)$ given by

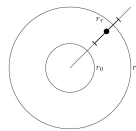
$$\log(Z_{N,\beta=2}^Q) = -I^Q N^2 + \frac{1}{2} N \log N + E^Q N + G \log N + F^Q + o(1),$$

as $N \rightarrow \infty$. For a large class of potentials Q , the first term I^Q was known for a long time, being given by the energy of the associated equilibrium measure. The coefficient $1/2$ of the order $N \log N$, as well as the entropy term E^Q , were known from recent work of Leblé and Serfaty [2], also for rather large classes of Q . The remaining terms in this expansion are new. The term G appears to depend only on whether the limiting spectrum is an annulus or a disc which, surprisingly, seems to not have been observed in the previous literature. Using some algebraic manipulation on the partition function, one can write the free energy for those models as

$$\log Z_N^Q = \log N! + \sum_{j=0}^{N-1} \log h_j, \quad h_j := \int_{\mathbb{C}} |z|^{2j} e^{-NQ(z)} dA(z).$$

If we're interested in the coefficients for the asymptotic expansion for the free energy, it is enough to understand the asymptotic of the summation of the norming constants.

Annulus



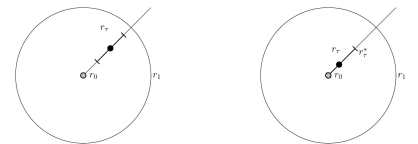
The first step in the asymptotic analysis is to change to radial variables $|z| = r$ and set $V_\tau(r) = q(r) - 2\tau \log(r)$ with $\tau(j) = j/N$. Then, we divide the integral using $\delta_N = \log(N)/\sqrt{N}$ for each h_j , as follows

$$\int_{|r-r_\tau|>\delta} e^{-NV_\tau(r)} 2r dr + \int_{|r-r_\tau|<\delta} e^{-NV_\tau(r)} 2r dr.$$

This is typical from the Laplace method - the first term gives the major contribution, the second corresponds to an error term, controlled in the asymptotic analysis. Then, to estimate the summation on $\log(h_j)$, we use the trapezoidal quadrature

$$\int_m^n f(x) dx = \sum_{j=m}^n f(j) - \frac{f(m) + f(n)}{2} - \sum_{k=1}^l \frac{B_{2k}}{(2k)!} (f^{(2k-1)}(n) - f^{(2k-1)}(m)) + R_l$$

Disc



An important observation made is that the asymptotic behavior of h_j depends on whether the degree j is sufficiently small or not. The case where j is big enough, the method is virtually the same as before - we are dealing with almost the same local contributions. The interesting part comes when $j > N^\epsilon$ for $0 < \epsilon < 1/5$. Set $r_\tau^* := r_\tau \cdot \log(N)$ and divide the integral again

$$\int_{r < r_\tau^*} e^{-NV_\tau(r)} 2r dr + \int_{r > r_\tau^*} e^{-NV_\tau(r)} 2r dr.$$

The first term gives the main contribution to the asymptotic and the second can be controlled. Near the center we should expect some other regimen, which is observed in the results. As mentioned, up to the N -ordered term, all is the same. The difference shows from the $\mathcal{O}(\log(N))$ and $\mathcal{O}(1)$ terms - related to the genus of the support. This difference appears to have not been considered before.

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C Paper: A Central Limit Theorem for intransitive dice

Published Paper in *ALEA*: “A Central Limit Theorem for intransitive dice”



A Central Limit Theorem for Intransitive Dice

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Abstract. Intransitive dice $D^{(1)}, \dots, D^{(\ell)}$ are dice such that $D^{(1)}$ has advantage when played against $D^{(2)}$, dice $D^{(2)}$ has advantage when played against $D^{(3)}$ and so on, up to $D^{(\ell)}$, which has advantage over $D^{(1)}$. This work has two main goals: to present (deterministic) results on the existence of general intransitive dice, and to derive a central limit theorem for the vector of normalized victories of a die against the next one in the list. As assumptions of this CLT, we assume that faces of a die are i.i.d. random variables and all dice are independent, but different dice may have distinct distributions associated with them, as well as they may have distinct numbers of faces. Exploiting this central limit theorem, we derive two major consequences. First, we are able to obtain first order exponential asymptotics for the number of ℓ -tuples of intransitive dice, when the number of faces of the dice grows. Second, we obtain a criterion to ensure that the asymptotic probability of observing intransitive dice is null. The criterion applies to many concrete situations, including all continuous distributions and many discrete ones.

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D Paper: Simulating Simple Random Walks With a Deck of Cards

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Simulating simple random walks with a deck of cards

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Abstract. When we want to simulate the realization of a symmetric simple random walk on $\mathbb{Z}^d$, we use $(2d)$ -side fair dice to decide to which neighbor it jumps at each step if $d \geq 2$ or we simply use a fair coin when $d = 1$. Assume that instead of using a dice or a coin we want to do a simulation using a well shuffled deck with K cards of each of the $2d$ suits. In the first step the probability of jumping to each neighbor is $(2d)^{-1}$, but from the second step it becomes biased. Of course if we continue performing this simulation, the total variation distance between its law and the law of the random walk will increase until all cards are used. In this paper we investigate the minimum number of cards $N = 2dK$ that a deck must contain so that the total variation distance between the law of a n -step simulation and the law of a n -step realization of the random walk is smaller than a chosen threshold $\varepsilon \in (0, 1)$. More generally, we prove that when $N = cn$ this distance converges, as $n \rightarrow \infty$, to a Gaussian profile which depends on $c \geq 2d$. Furthermore, our analysis shows that this Gaussian profile vanishes as $c \rightarrow \infty$, proving the convergence of a multivariate hypergeometric distribution to a multinomial distribution in total variation.

1 Introduction

The discrete time symmetric simple random walk (SSRW) on $\mathbb{Z}^d$ is one of the simplest examples of discrete time Markov chains: when at a vertex $x \in \mathbb{Z}^d$ it jumps to one of its $(2d)$ -equally-likely neighbors. Despite its simplicity, there are several studies involving random walks in biology, financial modeling, statistical physics and other sciences.

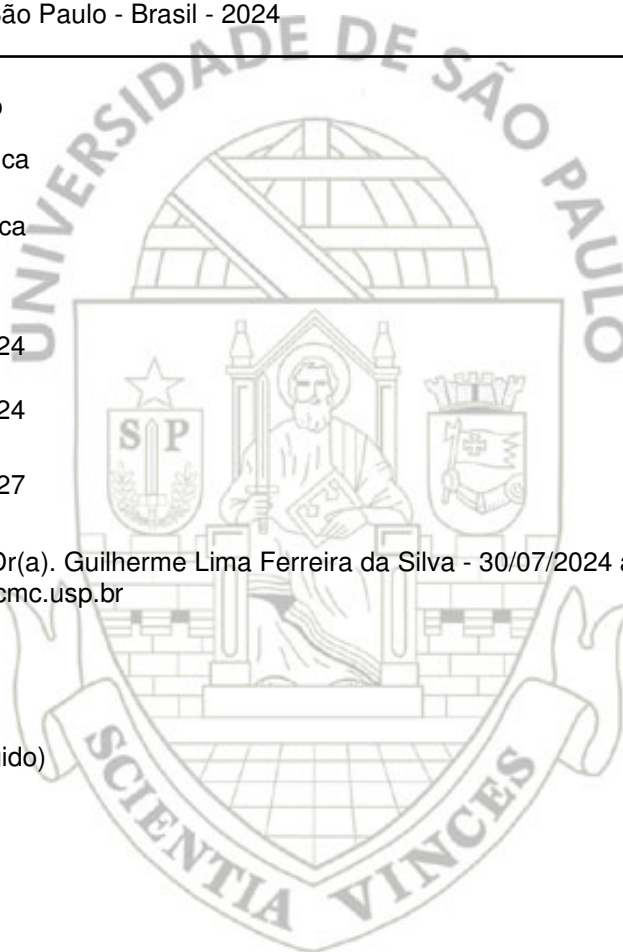
It is very common to use the SSRW on $\mathbb{Z}$, starting at the origin, to illustrate the Law of Large Numbers (LLN) and the Central Limit Theorem (CLT). The idea is that the position of the SSRW after n steps can be seen as the sum of n i.i.d. random variables taking values in $\{-1, 1\}$, and therefore, the average of this random variables converges almost surely to zero and the distribution of the aforementioned sum converges to a Gaussian distribution. Since we can associate i.i.d. random variables taking values in $\{-1, 1\}$ with independent fair coin tosses, this example can be easily explained to college students through a practical experiment, as well as Buffon's Needle Experiment, see Ramaley (1969). For the SSRW on $\mathbb{Z}^d$ one can adapt this experiment using d fair coins at each step, which would not be very practical if d is too large, or a $(2d)$ -sided dice.

In this work, we are interested in a practical experiment to simulate the SSRW on $\mathbb{Z}^d$, but using a well shuffled deck of cards instead of coins or dices. If $d = 1$, we can color half of the

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SMA5802-17/2	Equações Diferenciais Ordinárias	13/08/2024	29/11/2024	120	8	100	B	N	Concluída	S
SMA5738-17/4	Álgebra	10/03/2025	17/07/2025	120	8	96	A	N	Concluída	S
SMA5801-13/4	Medida e Integração	11/03/2025	17/07/2025	120	8	100	C	N	Concluída	S
SMA5839-19/8	Preparação Pedagógica	13/03/2025	17/07/2025	60	0	-	-	N	Matrícula cancelada	S
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